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United States Patent	12414764
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Handie; Andronica et al.

Medical device bending devices and methods of use

Abstract

Medical device bending devices and methods of use are disclosed herein. An embodiment includes a body, a handle in rotatable relationship with the body, and a head that is coupled with the handle in such a way that when the handle is rotated the head is drawn at least partially into the body and towards the handle. A terminal end of the body that is opposite of the handle and the head cooperate as a vise that bends opposing arms of a surgical staple from an inwardly angled configuration to a parallel configuration (or at angles therebetween) when a crown of the surgical staple is clamped in the vise.

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Appl. No.:	18/581239
Filed:	February 19, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240268819 A1	Aug. 15, 2024

Related U.S. Application Data

continuation parent-doc US 17170718 20210208 US 11931024 child-doc US 18581239
continuation parent-doc US 15978278 20180514 US 10912557 20210209 child-doc US 17170718
continuation parent-doc US 15644271 20170707 US 9993246 20180612 child-doc US 15978278

Publication Classification

Int. Cl.: **A61B17/068** (20060101); **A61B17/064** (20060101); **A61B17/10** (20060101); **A61B17/17** (20060101); A61B17/00 (20060101); A61B17/16 (20060101)

U.S. Cl.:

CPC **A61B17/068** (20130101); **A61B17/0642** (20130101); **A61B17/0644** (20130101); **A61B17/0682** (20130101); **A61B17/10** (20130101); **A61B17/17** (20130101); A61B2017/00477 (20130101); A61B2017/0645 (20130101); A61B17/16 (20130101)

Field of Classification Search

CPC: A61B (17/0644); A61B (17/068); A61B (17/8863)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This non-provisional application is a continuation of and claims the benefit and priority of U.S. application Ser. No. 17/170,718, filed on Feb. 8, 2021, titled MEDICAL DEVICE BENDING DEVICES AND METHODS OF USE, now U.S. Pat. No. 11,931,024, which is a continuation of U.S. application Ser. No. 15/978,278, filed on May 14, 2018, titled MEDICAL DEVICE BENDING DEVICES AND METHODS OF USE, now U.S. Pat. No. 10,912,557, which is a continuation of and claims the benefit and priority of U.S. application Ser. No. 15/644,271 filed on Jul. 7, 2017, titled MEDICAL DEVICE BENDING DEVICES AND METHODS OF USE, now U.S. Pat. No. 9,993,246, which are hereby incorporated by reference herein in their entireties, including all references and appendices cited therein.

FIELD OF THE PRESENT DISCLOSURE

(1) Embodiments of the present disclosure are directed to surgical instruments, and more particularly, to but not by limitation, to medical device bending devices/apparatuses that are configured to bend, for example, surgical staples.

SUMMARY

(2) According to some embodiments, the present disclosure is directed to a device, comprising: a body; a handle in rotatable relationship with the body; and a head that is coupled with the handle in such a way that when the handle is rotated, the head is drawn partially into the body and towards the handle, wherein a terminal end of the body that is opposite of the handle and the head cooperate as a vise, the vise clamping a crown of a surgical staple in such a way that ends of opposing arms of the surgical staple that extend from the crown move away from one another.

(3) According to some embodiments, the present disclosure is directed to a method, comprising: inserting a surgical staple in a device, the device comprising: a body; a handle in rotatable relationship with the body; and a head that is coupled with the handle in such a way that when the handle is rotated, the head is drawn partially into the body and towards the handle, wherein a terminal end of the body that is opposite of the handle and the head cooperate as a vise that bends opposing arms of a surgical staple from the inwardly angled configuration to a parallel configuration when a crown of the surgical staple is clamped in the vise; and rotating the handle to transition the surgical staple from the inwardly angled configuration to a parallel configuration.

(4) According to some embodiments, the present disclosure is directed to a kit, comprising: a surgical staple in an initial inwardly angled configuration (a shape memory material staple); and a device comprising: a body; a handle in rotatable relationship with the body; and a head that is coupled with the handle in such a way that when the handle is rotated, the head is drawn partially into the body and towards the handle, wherein a terminal end of the body that is opposite of the handle and the head cooperate as a vise that bends opposing arms of a surgical staple from the inwardly angled configuration to a parallel configuration when a crown of the surgical staple is clamped in the vise.

(5) The kit can also include a pilot hole jig and pilot hole drill for creating pilot holes in a patient. The pilot hole jig has jig cylinders spaced apart from one another so as to create spacing between the pilot holes that correspond to a spacing of ends of the surgical staple when in the parallel configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed disclosure, and explain various principles and advantages of those embodiments.

(2) The methods and systems disclosed herein have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

(3) FIG. 1 is a cross-sectional view of an example device constructed in accordance with the present disclosure.

(4) FIG. 2 includes plan views of the example device of FIG. 1 in combination with a surgical staple, where the surgical staple is in an initial, inwardly bent configuration and a parallel

configuration.

(5) FIG. 3 is a perspective view of a head assembly of the device of FIG. 1.

(6) FIG. 4 illustrates plan views of the body of the example device of FIG. 1.

(7) FIG. 5 is a perspective view of a kit comprising the device of FIG. 1 in combination with a pilot hole jig tool and a pilot hole drill bit.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(8) The present disclosure is directed, in some embodiments, to surgical apparatuses, devices, kits, and methods of use that allow a user to bend surgical staples from a first configuration to a second configuration. When in the second configuration, the surgical staple can be inserted into a patient to secure structures together. Example structures include bone, skin, cartilage, and other similar structures that would be known to one of ordinary skill in the art with the present disclosure before them.

(9) In some embodiments, an example surgical device comprises a body that is configured to be gripped by a user. The surgical device also comprises a handle in rotatable relationship with the body. In various embodiments, the surgical device comprises a head that is coupled with the handle in such a way that when the handle is rotated, the head is drawn partially into the body and towards the handle.

(10) In one or more embodiments, the handle is rotatably coupled with the head such that turning of the handle results in upward (e.g., towards the handle) movement of the head.

(11) With respect to the bending of a surgical staple, the body and head of the surgical device form a vise that bends a crown of the surgical staple causing arms of the surgical staple to transition from an inwardly angled configuration to a parallel configuration. That is, the surgical staple is transitioned when the handle is turned, which results in the head being drawn towards a terminal end of the body, thereby pinching or compressing the crown of the staple.

(12) Stated otherwise, the vise clamps a crown of a surgical staple in such a way that ends of opposing arms of the surgical staple that extend from the crown move away from one another.

(13) The devices of the present disclosure are adapted to bending medical devices, such as surgical staples that are manufactured from shape memory alloy material. When the staple is removed from an example device after insertion of the surgical staple into the patient, the surgical staple returns to its initial configuration due to its shape memory material properties.

(14) These and other advantages of the present disclosure are described in greater detail infra with respect to the collective drawings FIGS. 1-5.

(15) FIG. 1 is a cross sectional view of an example device **100**, such as a surgical staple bending apparatus. The device **100** generally comprises a handle assembly **102**, a body assembly **104**, and a head assembly **106**. In some embodiments, the handle assembly **102** comprises a handle **108** and a shaft **110**. In various embodiments, the shaft **110** has a threaded end **112** that is configured for threaded engagement of the head assembly **106**, as will be discussed in greater detail infra.

(16) According to some embodiments, the shaft **110** is inserted through the body assembly **104** allowing the handle **108** to rotate about a first end or handle end of the body assembly **104**.

(17) Turning to the body assembly **104**; it comprises a body **114** and protrusion **116** that forms a first portion of a vise **122**. In some embodiments, the body **114** comprises a handle end **118** and a terminal end **120** that is opposite to the handle end **118**. In some embodiments, the body **114** has curved surfaces that allow a user to easily grip and/or manipulate the body **114** (and by extension the device **100**).

(18) In accordance with the present disclosure, the body **114** comprises a linear passage **124** that provides a pathway from the handle end **118** to the terminal end **120**. The linear passage **124** can comprise a cylindrical aperture that is sized to receive the shaft **110** of the handle assembly **102**. The shaft **110** can freely rotate within the linear passage **124** based on rotation of the handle **108**.

(19) Also, the body **114** comprises a recess **126** that receives a portion of the head assembly **106**.

(20) Referring briefly to FIG. 4, the body **114** comprises a channel **127** on the handle end **118**, as

well as a channel **129** on the terminal end **120**. The device **100** can comprise either or both channels **127** and **129**. Functionally, the channels provide mating interfaces that receive a portion of a surgical staple and allow the device **100** to be used as a tool for imparting percussive force to the surgical staple. These features are described in greater detail herein.

(21) In one or more embodiments, the protrusion **116** comprises a hemispherical protrusion that at least partially overhangs a terminal end edge **128**. The protrusion **116** also extends above an upper surface **117** of the body **114**.

(22) The head assembly **106**, in some embodiments, comprises a dowel **130** and a block **132**. The dowel **130** is configured for sliding insertion into the recess of the body **114**. The dowel **130** comprises, in various embodiments, a threaded receiver **134**. The threaded end **112** of the shaft **110** engages with the threaded receiver **134**.

(23) Rotation of the handle **108** in a first direction causes the threaded end **112** to drive into the threaded receiver **134**, causing the dowel **130** to be drawn inwardly into the recess **126** (towards the handle **108**) and the block **132** towards the terminal end **120** of the body **114**. Rotation of the handle **108** in a second direction that is opposite of the first direction causes the threaded end **112** to be extracted from the threaded receiver **134**, allowing the dowel **130** and block **132** to translate away from the handle **108**.

(24) Referring briefly to FIG. 3, according to some embodiments the block **132** functions as a second portion of the vise **122**. In some embodiments, the block **132** comprises a flange **136** that extends from the dowel **130**. Disposed on the flange **136** is a staple interface **138**. The staple interface **138** comprises a groove **140** that is formed into a portion of the staple interface **138** at a terminal end of the flange **136**.

(25) In some embodiments, the groove **140** comprises rounded or arcuate ends **142** and **144**.

(26) Referring to FIGS. 1-3 collectively, the protrusion **116** of the body **114** and the groove **140** are coplanar with one another (as best illustrated in FIG. 1).

(27) A surgical staple **146** can be cradled and held in place within the groove **140**, as illustrated in FIG. 1, where the surgical staple is illustrated in phantom dotted line so that the groove **140** is viewable. More specifically, and turning to FIG. 3, the surgical staple **146** comprises a crown **148** and two arms **150** and **152**. Each of the arms **150** and **152** has an end. For example, arm **150** has an end **154** and the arm **152** has an end **156**.

(28) Turning now to FIG. 2, the surgical staple **146** is initially in an inwardly angled configuration CI where the crown **148** is arcuate and the ends **154** and **156** bend inwardly towards one another.

(29) When surgical staple **146** is placed into the groove **140** in its inwardly angled configuration CI, the arcuate shape of the crown **148** allows for a space or gap **158** to exist between the crown **148** and a substantially linear portion **160** of the groove **140**.

(30) This space **158** allows the crown **148** to deflect towards the substantially linear portion **160** of the groove **140** when the surgical staple **146** is clamped in the vise **122**. Moreover, when the crown **148** is clamped in the vise **122**, the crown **148** will begin to straighten into a linear configuration from its original arcuate configuration. Clamping of the crown **148** of the surgical staple **146** will cause the ends **154** and **156** to splay and/or move away from one another. This can result in the arms **150** and **152** transitioning to a substantially parallel configuration C2 (see FIG. 2). To be sure, not all embodiments and use cases require the surgical staple **146** to be in the substantially parallel configuration C2. Therefore, transitioning the surgical staple from the inwardly angled configuration CI to an intermediate configuration that is less than the substantially parallel configuration C2 is also contemplated.

(31) According to some embodiments, and turning to FIG. 3, the block **132** comprises a pair of protrusions **162** and **164** that extend laterally from the block **132**. These protrusions are disposed behind the groove **140**. When the surgical staple **146** is coupled with the groove **140**, a portion of the surgical staple **146** abuts or contacts the protrusions **162** and **164**. These protrusions **162** and **164** are optional in some embodiments. The protrusions can be used to support or stabilize the

surgical staple **146** during bending and/or insertion.

(32) The device **100** can be comprised of any suitable material that is capable of being used in a surgical operation. Some embodiments can be manufactured from materials that are amenable to sterilization and other embodiments may be manufactured from materials that allow for sterilization and reuse.

(33) In operation, a user can utilize the device **100** to bend the surgical staple **146**. The user can ensure that a space is created between the terminal end **120** of the body **114** and the groove **140** of the block **132** of the head assembly **106** so that the surgical staple **146** can be inserted into the groove **140** of the block **132**. Once in position, the user can begin to turn the handle **108** which rotates the shaft **110** causing the threaded end **112** to drive into the threaded receiver **134** of the dowel **130**. This in turn, causes the dowel **130** to translate upwardly into the recess **126** of the body **114** towards the handle **108**. Upward movement of the dowel **130** causes the block **132** to be drawn into proximity of the protrusion **116** on the body **114**. The surgical staple **146** begins to bend when the crown **148** is clamped between the protrusion **116** of the body **114** and the groove **140** of the block **132**. In more detail, as the block **132** brings the crown **148** into contact with the protrusion **116**, the gap **158** between the crown **148** and the groove **140** begins to reduce in size. As the crown **148** is compressed between the block **132** and the protrusion **116**, the ends **154** and **156** begin to move away from one another.

(34) When the handle **108** has been turned enough so that it contacts the handle end **118** of the body **114**, the crown **148** is approximately compressed flatly (or is in mating contact in some embodiments) against the substantially linear portion **160** of the groove **140**. In this position, the arms **150** and **152** of the surgical staple **146** are approximately parallel (or completely parallel) with one another. Stated otherwise, the compression of the crown **148** of the surgical staple **146** between the protrusion **116** and block **132** causes the arms **150** and **152** to splay outwardly from the inwardly angled configuration C1 to the parallel configuration C2.

(35) In some embodiments, the shaft **110** is sized in length such that when the handle **108** contacts the handle end **118** of the body **114**, the crown **148** of the surgical staple **146** is compressed between the block **132** and the protrusion **116** to place the arms **150** and **152** in parallel relationship to one another. Contact between the handle **108** and the handle end **118** of the body **114** then prevents further rotation of the handle **108**, which could lead to over compression of the crown **148** of the surgical staple **146** (e.g., over-bending of the surgical staple **146**). Thus, the length of the shaft **110** can vary such that the surgical staple **146** cannot be bent further than required. This could be based on spacing between pilot holes that receive the arms **150** and **152** of the surgical staple **146**. The details of creating the pilot holes are described in greater detail infra.

(36) When inserted into the patient, the device **100** can be decoupled from the surgical staple **146** by rotating the handle **108** in an opposing direction from the direction used to bend the surgical staple **146**. When the handle **108** is rotated in the reverse direction, the shaft **110** rotates so as to begin to back out of the threaded receiver **134** of the dowel **130**. This causes the dowel **130** to move away from the handle **108**, which in turn causes the block **132** and the protrusion **116** to separate. When the block **132** and the protrusion **116** are sufficiently separated, the surgical staple **146** is removed from the block **132**. In accordance with the present disclosure, the surgical staple **146** can be driven further into the patient using the body **114** of the device **100**.

(37) For example, the handle **108** and head assembly **106** can be removed from the body **114**. Either of the channels **127** or **129** can be mated with the crown **148** of the surgical staple **146**. That is, the crown **148** fits into whichever channel is selected. The user then applies a percussive force to an opposing end of the body **114** that is not engaged with the crown **148** driving the surgical staple **146** further into the patient, such as into a bone of the patient.

(38) FIG. 5 illustrates an example kit that comprises a device **100** of the present disclosure in combination with a pilot hole device **202** and pilot drill bit **204**. In some embodiments, the pilot hole device **202** comprises a handle **206** and pilot hole jig **208**. The pilot hole jig **208** comprises jig

cylinders **210** and **212** that are spaced apart from one another. The pilot drill bit **204** is used to create pilot holes in a patient structure, such as a bone. That is, when the pilot hole jig **208** is positioned on the patient structure, the pilot drill bit **204** is inserted into each of the jig cylinders **210** and **212** respectively. The pilot drill bit **204** is used to ream out pilot holes in the patient structure.

(39) As mentioned above, spacing between the jig cylinders **210** and **212** is set so as to correspond to spacing between arms **150** and **152** (or ends **154** and **156**) of the surgical staple **146**.

(40) After creation of the pilot holes, the surgical staple **146** is bent in the device **100** according to the method described supra. The arms **150** and **152** of the surgical staple **146** are then inserted into the pilot holes.

(41) In some embodiments, the kit further comprises one or more spikes **214** and **216** that are used to fill pilot holes after creation with the pilot hole jig **208** and the pilot drill bit **204**.

(42) Example uses for the devices of the present disclosure include, but are not limited to orthopedic procedures. For example, the devices of the present disclosure can be used to fuse or secure together separated segments of a bone, such as two sections of a broken metatarsal bone. The user can place the pilot hole jig **208** onto mating ends of the broken metatarsal bone and pilot holes are drilled. A surgical staple is then inserted into the pilot holes. It will be understood that the surgical staple is bent prior to insertion into the pilot holes. In some embodiments, the surgical staple is manufactured from a resiliently biased or shape memory material such as Nitinol™. The material properties of the surgical staple cause the surgical staple to return to its initial configuration (e.g., inwardly bent configuration CI, see FIG. 2), which draws the mating ends of the broken metatarsal bone toward one another and holds the mating ends together to promote fusion.

(43) While this technology is susceptible of embodiment in many different forms, there is shown in the drawings and has been described in detail several specific embodiments with the understanding that the present disclosure is to be considered as an exemplification of the principles of the technology and is not intended to limit the technology to the embodiments illustrated.

(44) Although the terms first, second, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not necessarily be limited by such terms. These terms are only used to distinguish one element, component, region, layer or section from another element, component, region, layer or section. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the present disclosure.

(45) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be necessarily limiting of the disclosure. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “includes” and/or “comprising,” “including” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(46) Example embodiments of the present disclosure are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the present disclosure. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, the example embodiments of the present disclosure should not be construed as necessarily limited to the particular shapes of regions illustrated herein, but are to include deviations in shapes that result, for example, from manufacturing.

(47) Any and/or all elements, as disclosed herein, can be formed from a same, structurally continuous piece, such as being unitary, and/or be separately manufactured and/or connected, such

as being an assembly and/or modules. Any and/or all elements, as disclosed herein, can be manufactured via any manufacturing processes, whether additive manufacturing, subtractive manufacturing and/or other any other types of manufacturing. For example, some manufacturing processes include three dimensional (3D) printing, laser cutting, computer numerical control (CNC) routing, milling, pressing, stamping, vacuum forming, hydroforming, injection molding, lithography and/or others.

(48) Any and/or all elements, as disclosed herein, can include, whether partially and/or fully, a solid, including a metal, a mineral, a ceramic, an amorphous solid, such as glass, a glass ceramic, an organic solid, such as wood and/or a polymer, such as rubber, a composite material, a semiconductor, a nano-material, a biomaterial and/or any combinations thereof. Any and/or all elements, as disclosed herein, can include, whether partially and/or fully, a coating, including an informational coating, such as ink, an adhesive coating, a melt-adhesive coating, such as vacuum seal and/or heat seal, a release coating, such as tape liner, a low surface energy coating, an optical coating, such as for tint, color, hue, saturation, tone, shade, transparency, translucency, non-transparency, luminescence, anti-reflection and/or holographic, a photo-sensitive coating, an electronic and/or thermal property coating, such as for passivity, insulation, resistance or conduction, a magnetic coating, a water-resistant and/or waterproof coating, a scent coating and/or any combinations thereof.

(49) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. The terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an idealized and/or overly formal sense unless expressly so defined herein.

(50) Furthermore, relative terms such as “below,” “lower,” “above,” and “upper” may be used herein to describe one element's relationship to another element as illustrated in the accompanying drawings. Such relative terms are intended to encompass different orientations of illustrated technologies in addition to the orientation depicted in the accompanying drawings. For example, if a device in the accompanying drawings is turned over, then the elements described as being on the “lower” side of other elements would then be oriented on “upper” sides of the other elements. Similarly, if the device in one of the figures is turned over, elements described as “below” or “beneath” other elements would then be oriented “above” the other elements. Therefore, the example terms “below” and “lower” can, therefore, encompass both an orientation of above and below.

(51) The description of the present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the present disclosure in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the present disclosure. Exemplary embodiments were chosen and described in order to best explain the principles of the present disclosure and its practical application, and to enable others of ordinary skill in the art to understand the present disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

(52) While various embodiments have been described above, it should be understood that they have been presented by way of example only, and not limitation. The descriptions are not intended to limit the scope of the technology to the particular forms set forth herein. Thus, the breadth and scope of a preferred embodiment should not be limited by any of the above-described exemplary embodiments. It should be understood that the above description is illustrative and not restrictive. To the contrary, the present descriptions are intended to cover such alternatives, modifications, and equivalents as may be included within the spirit and scope of the technology as defined by the appended claims and otherwise appreciated by one of ordinary skill in the art.

(53) The scope of the technology should, therefore, be determined not with reference to the above description, but instead should be determined with reference to the appended claims along with their full scope of equivalents.

Claims

1. A device, comprising: a body; a handle; and a head that is coupled with a handle in such a way that when the handle is rotated the head is drawn at least partially into the body and towards the handle, wherein a terminal end of the body has a hemispherical protrusion that is opposite of the handle and the head cooperate as a vise that bends opposing arms of a surgical staple when a crown of the surgical staple is clamped in the vise.
 2. The device according to claim 1, further comprising a threaded shaft that couples the handle and the head.
 3. The device according to claim 2, wherein the body comprises a linear passage that receives the threaded shaft.
 4. The device according to claim 3, wherein an end of the threaded shaft is threaded onto the head.
 5. The device according to claim 1, wherein the body comprises a recess that receives a dowel of the head.
 6. The device according to claim 1, wherein the head comprises a dowel and a block, the block comprising a second portion of the vise.
 7. The device according to claim 1, wherein the hemispherical protrusion overhangs a terminal end edge.
 8. The device according to claim 6, wherein the head comprises lateral protrusions that stabilize the surgical staple.
 9. The device according to claim 1, wherein the threaded shaft comprises a length such that when the handle is rotated to draw the head upwardly towards the handle, the handle contacts the body to prevent over bending of the surgical staple beyond the parallel configuration.
 10. The device according to claim 1, wherein a block of the head comprises a groove having arcuate ends, the groove receiving a portion of the surgical staple.
 11. The device according to claim 1, further comprising a first channel extending along a handle end of the body.
 12. The device according to claim 11, further comprising a second channel extending along the terminal end of the body.
 13. A method, comprising: inserting a surgical staple in a device, the device comprising: a body; a handle; and a head that is coupled with a handle in such a way that when the handle is rotated the head is drawn partially into the body and towards the handle, wherein a terminal end of the body includes a protrusion that is opposite of the handle and the head cooperate as a vise that bends opposing arms of a surgical staple when a crown of the surgical staple is clamped in the vise; and rotating the handle to transition the surgical staple from the inwardly angled configuration to a parallel configuration.
 14. The method according to claim 13, further comprising inserting the surgical staple into a patient.
 15. The method according to claim 13, further comprising: aligning the at least one channel with a crown of the surgical staple; and applying percussive force to the body to drive the surgical staple into a patient.
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